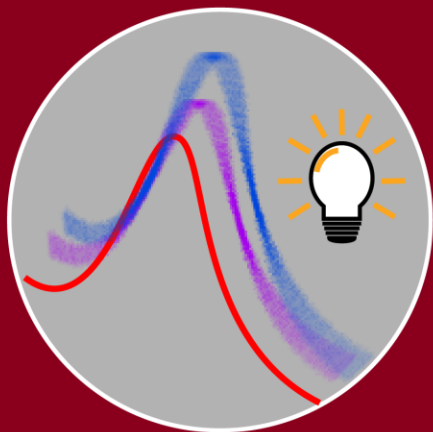
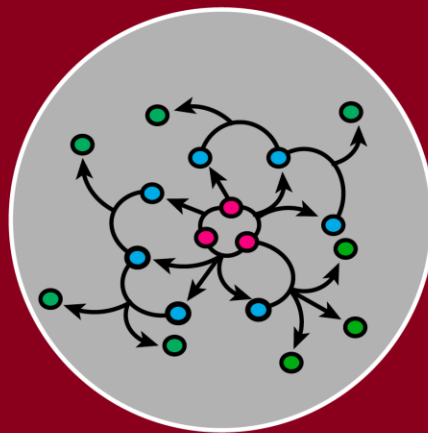
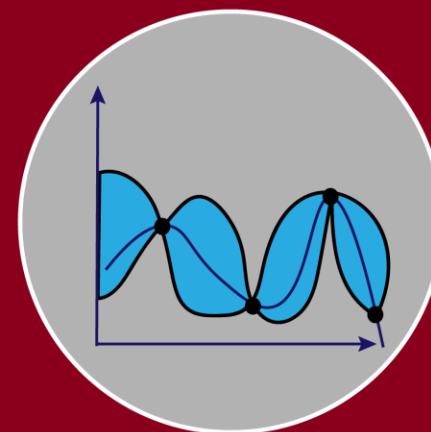




**data efficient reaction
screening & discovery**



**reaction network
exploration**

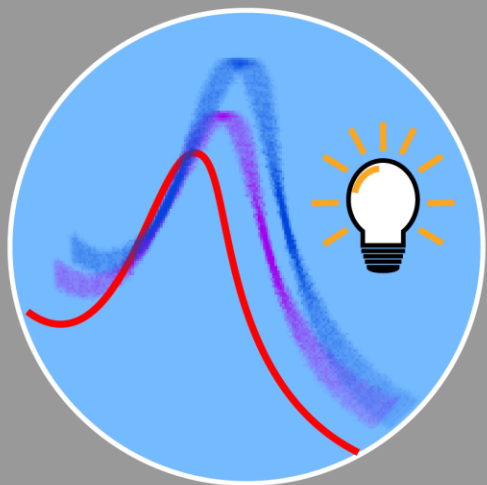


**ML & BO for
experiment design**

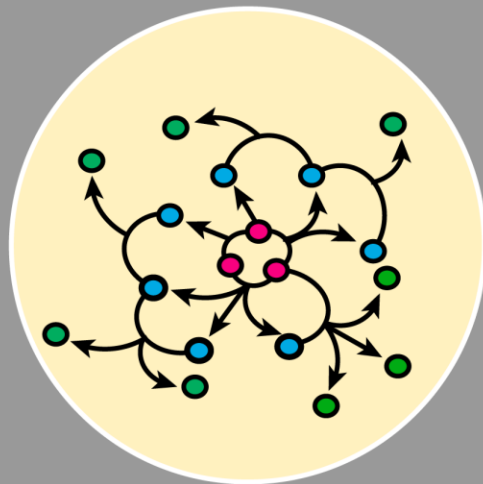
**Combining chemical theory and machine learning
techniques to accelerate reaction discovery**

Research overview

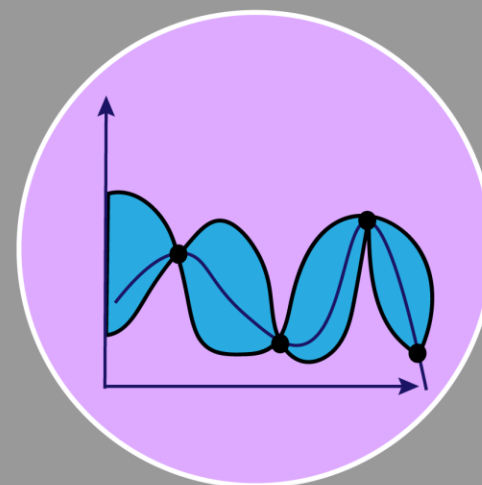
**Combining chemical theory and machine learning techniques
to accelerate chemical discovery**



**data efficient reaction
screening & discovery**



**reaction network
exploration**

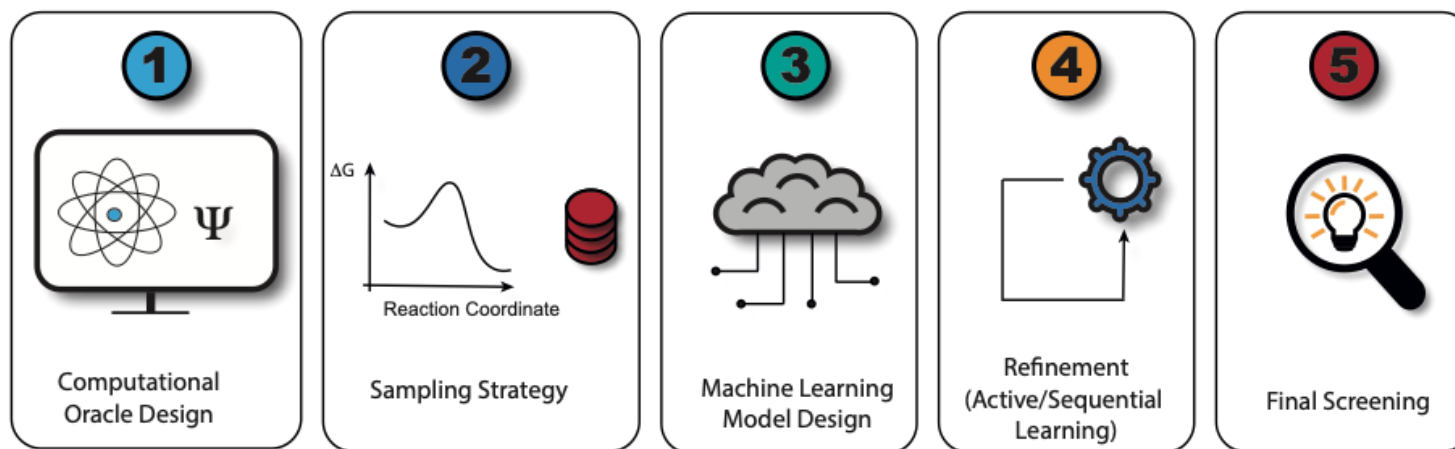


**ML & BO for
experiment design**

Data efficient reaction screening & discovery

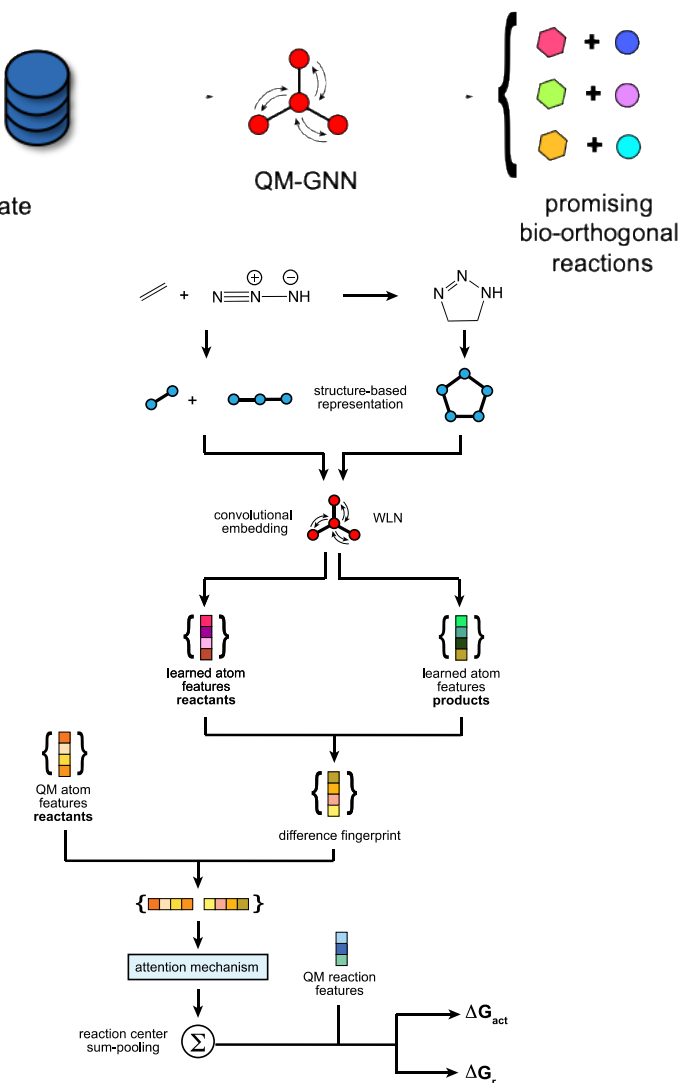
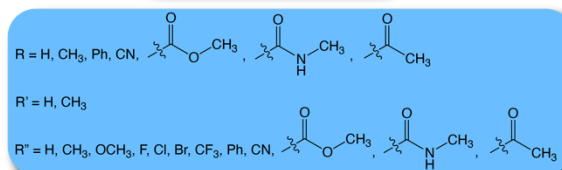
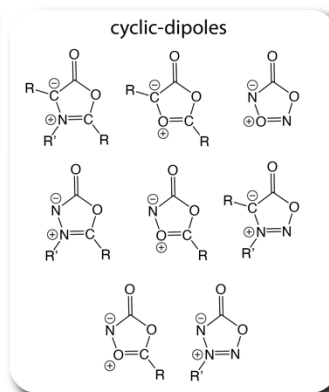
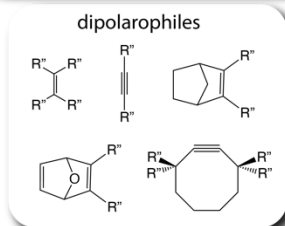
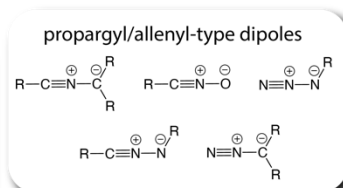
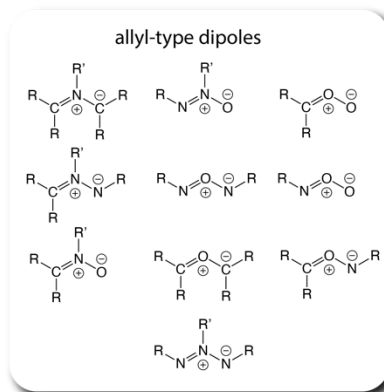
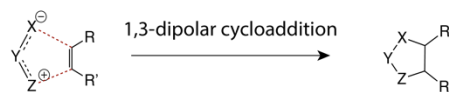
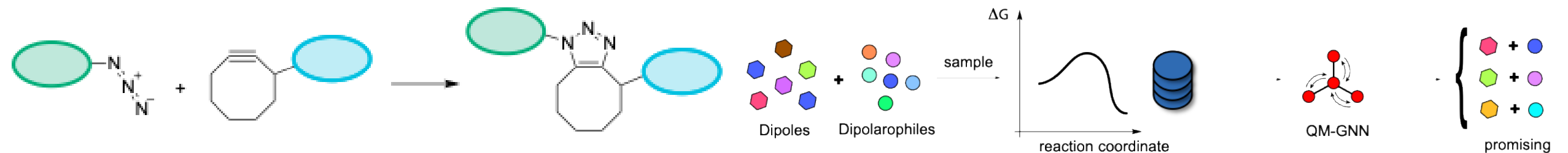
Conventional machine learning algorithms work really well with tens of thousands of data points...

... but in the data limited regime, they perform terribly



hybrid computational chemistry – machine learning workflows

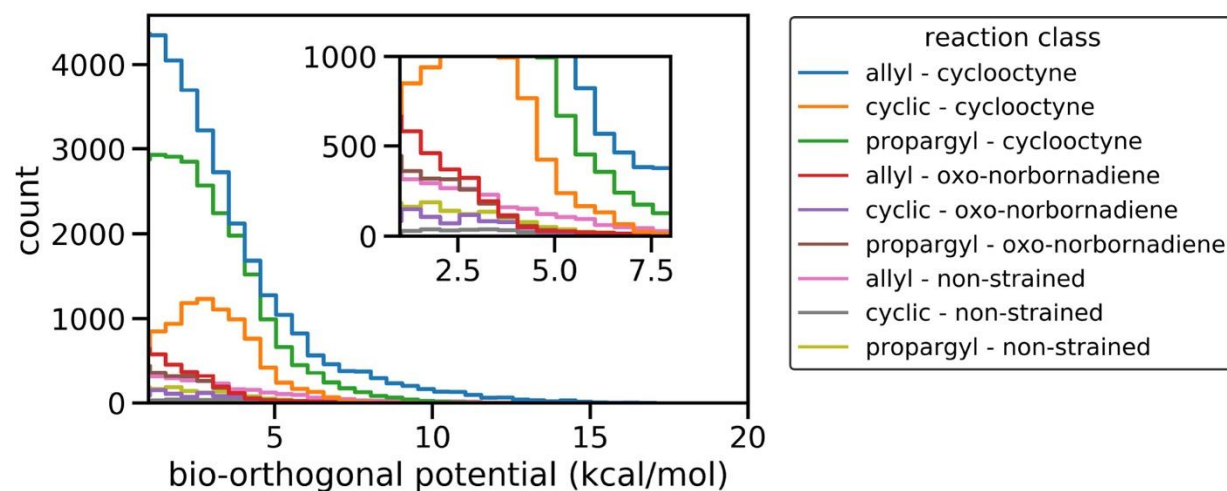
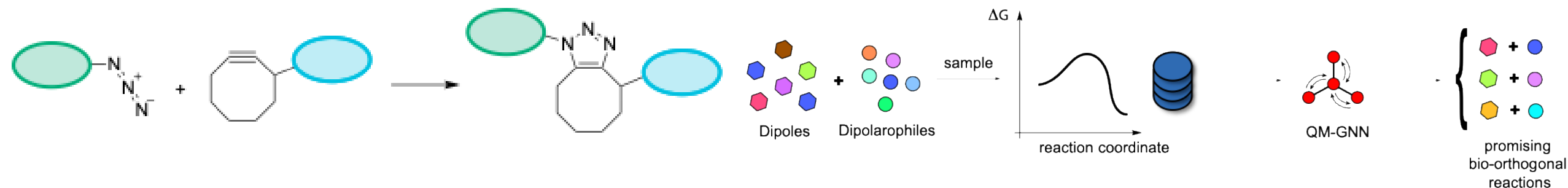
An early example – screening for bioorthogonal click reactions



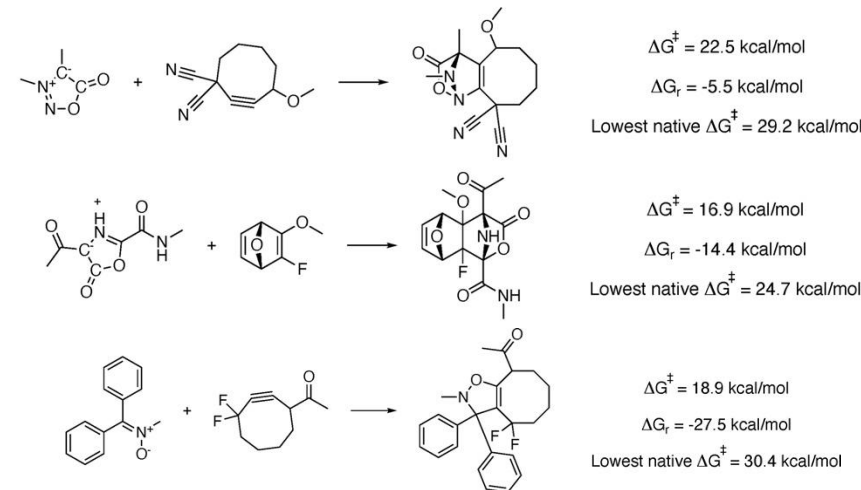
T. Stuyver, K. Jorner, C. W. Coley, *Sci. Data* **2023**, *10*, 66.

T. Stuyver, C. W. Coley, *Chem. Eur. J.* **2023**, e202300387.

An early example – screening for bioorthogonal click reactions

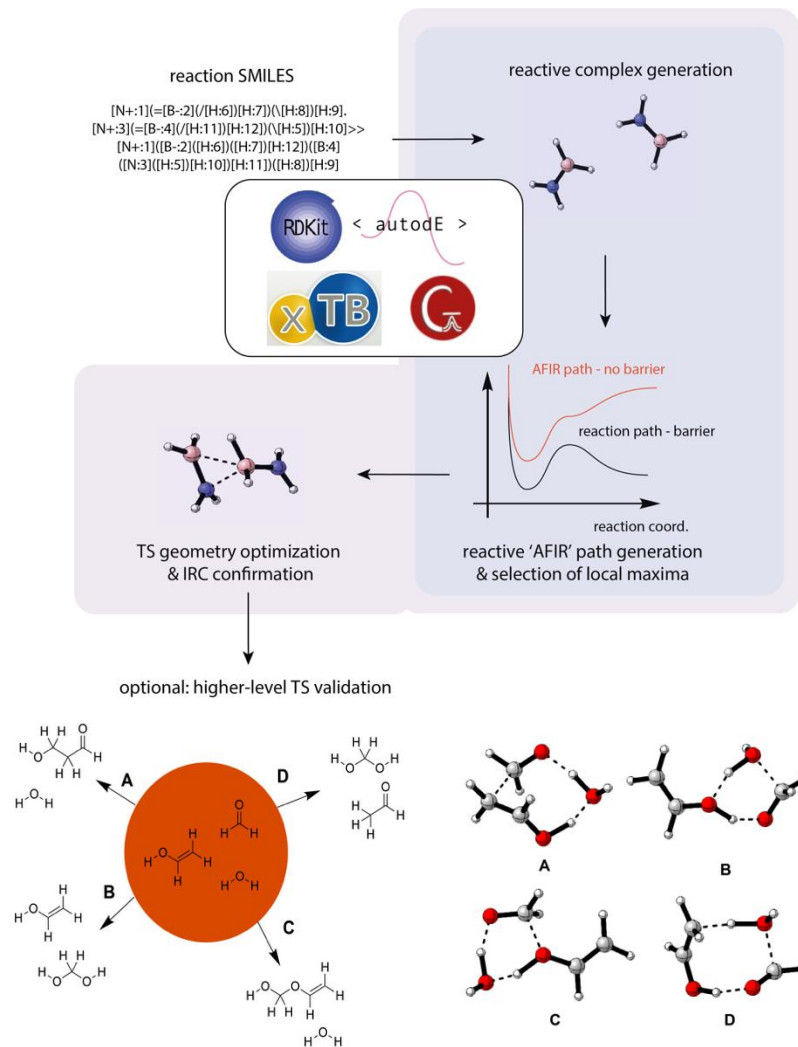


120.000+ reactions with suitable reactive properties



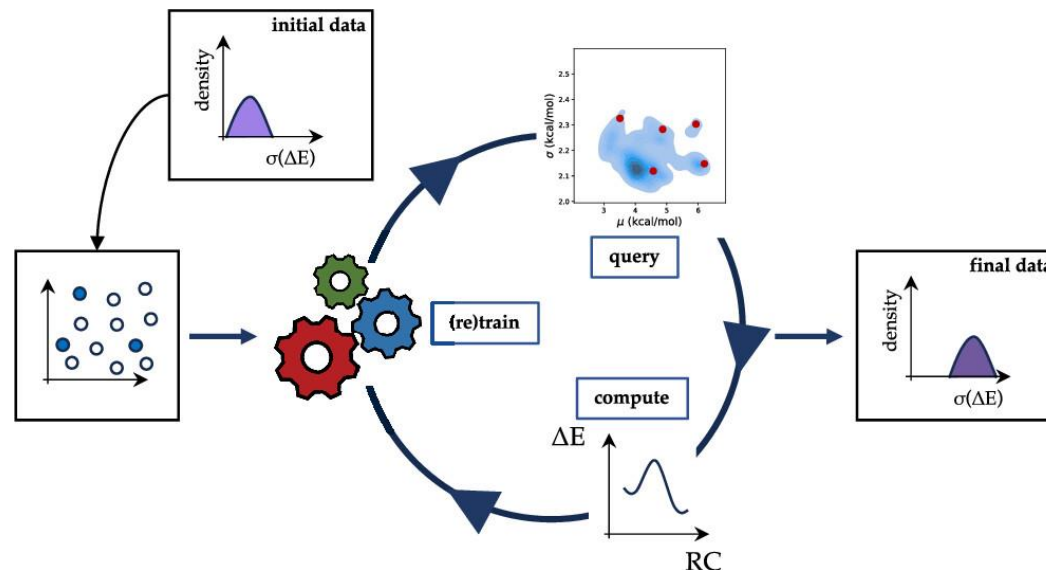
Methodological advances...

Rapid, automated localization of TS guesses



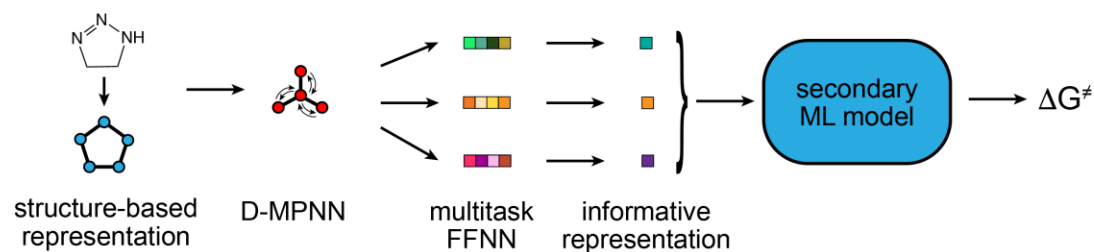
T. Stuyver, *J. Comput. Chem.* **2024**, 45, 2308-2317.

Constructing representative density functional theory benchmarking sets through active learning



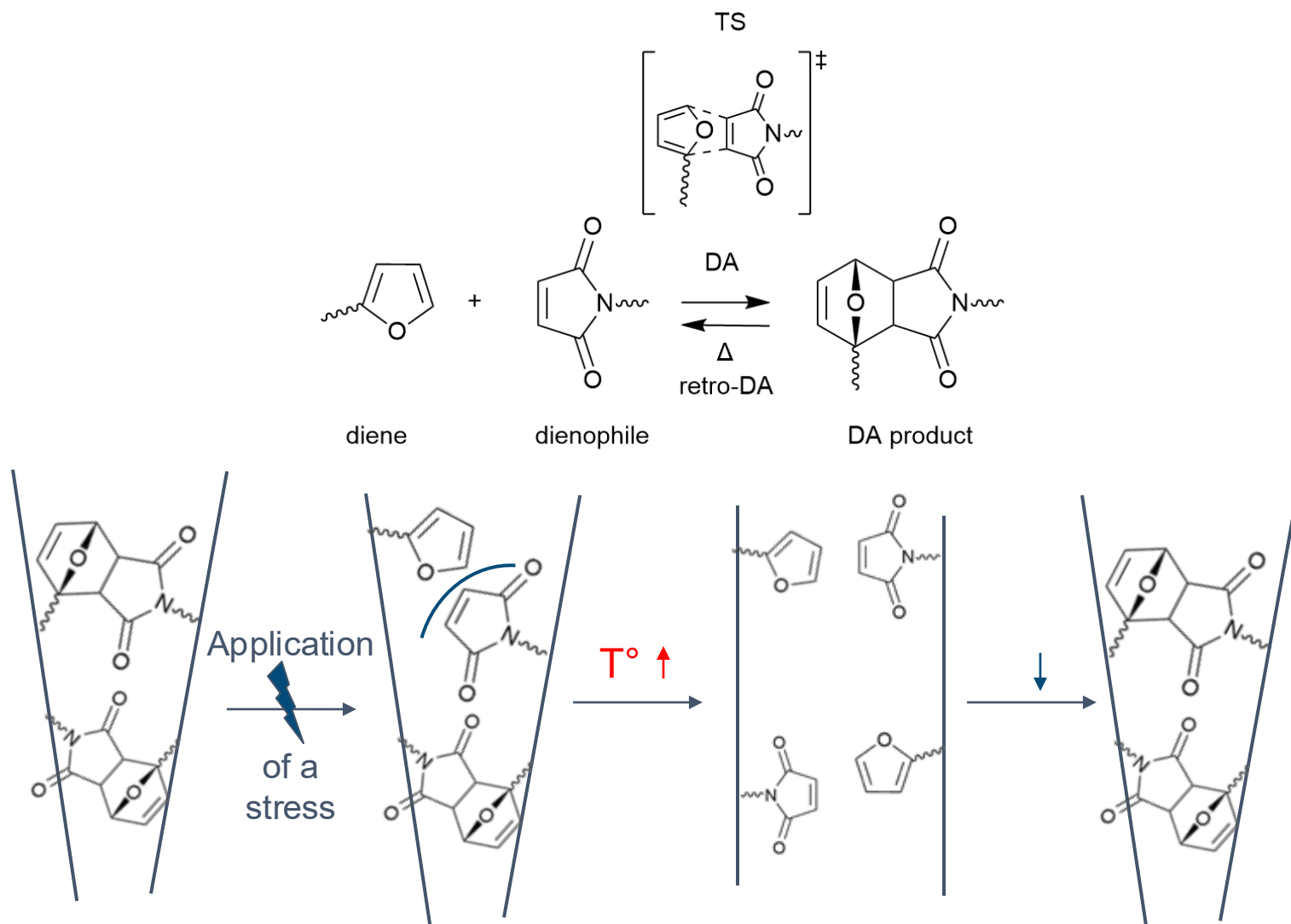
J. E. Alfonso-Ramos, C. Adamo, E. Brémond, T. Stuyver, *J. Chem. Theory Comput.* **2025**, 21, 1752-1761.

Using meaningful intermediate representations to improve the data efficiency of ML models

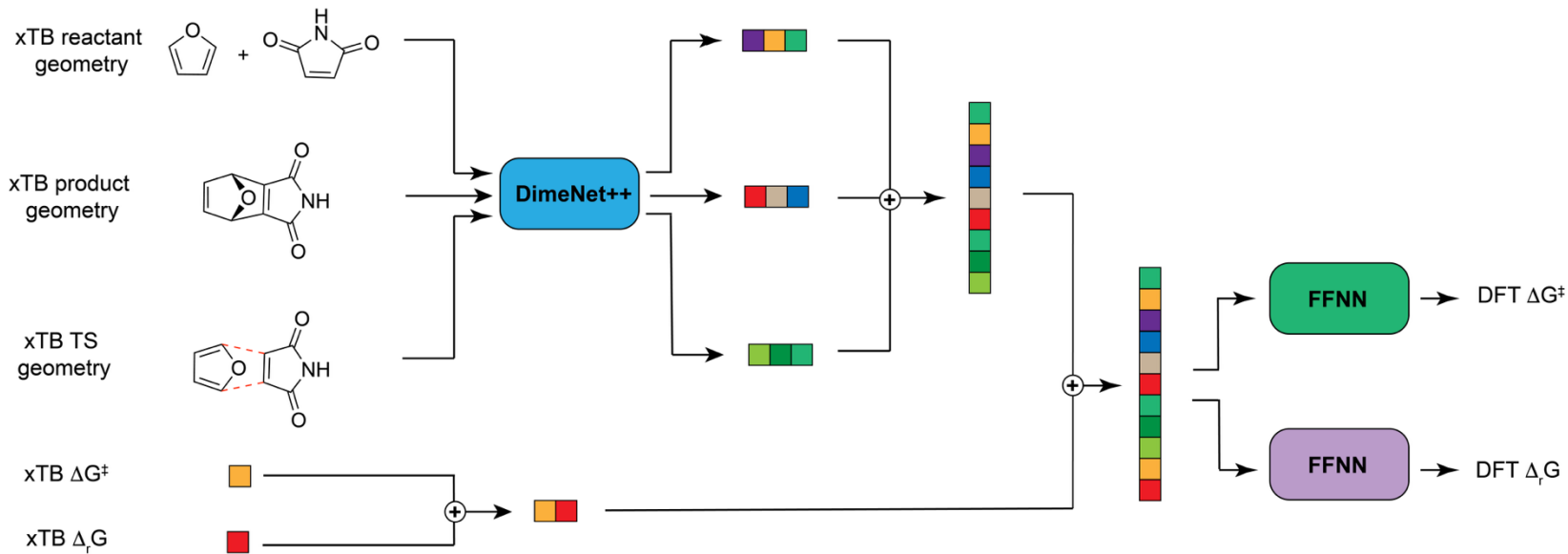
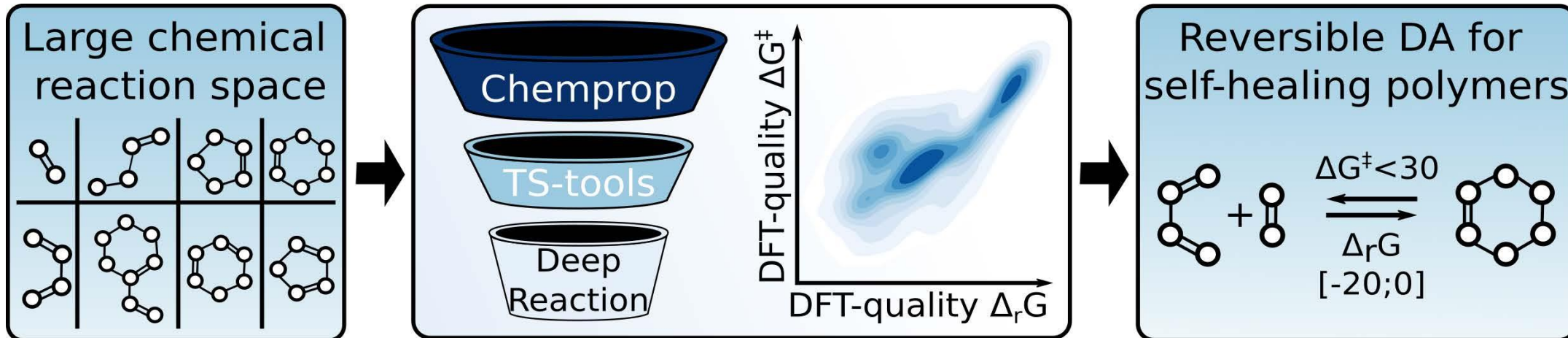


J. E. Alfonso-Ramos, R. Neeser, T. Stuyver *Digit. Discov.* **2024**, 3, 919-931.

... and applications – self-healing polymers

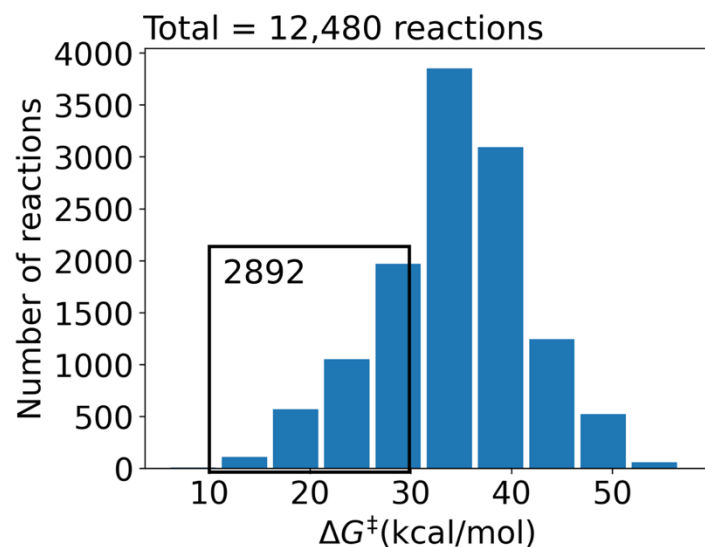


... and applications – self-healing polymers

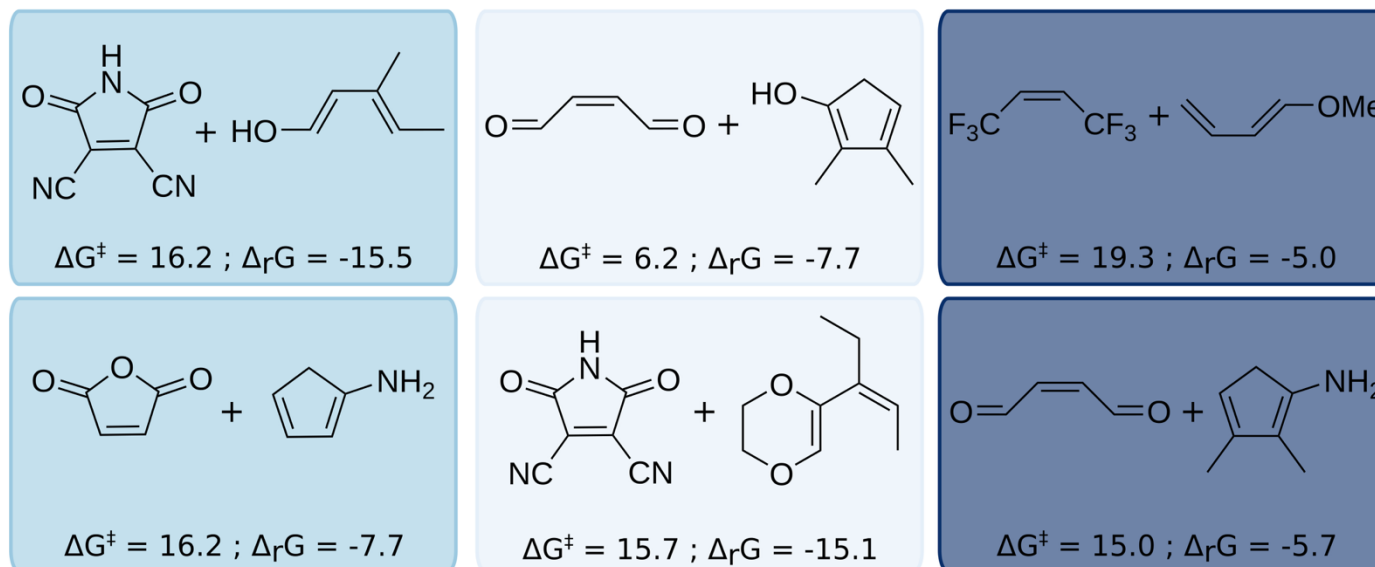


... and applications – self-healing polymers

A



B



Acknowledgements



Javier E. ALFONSO-
RAMOS



Bowen DENG



Maxime FERRER



Basile PARMELLI



Emilien CHEN



Carlo ADAMO



Ilaria CIOFINI



Alistar OTTOCHIAN

Thank you for your attention!

